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LABORATORY OF DATA ENGINEERING

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# Web Application for DPCfam and DPCstruct Data Exploration

Report 5 (9 Feb - 14 Feb 2026) : Web Development  
(DPCFam Backend)

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# 1 Introduction

This report is the fifth in a sequence describing the completion of weekly assigned tasks during our internship. Focusing on the DPCFam dataset(Russo et al., 2022), we present the backend methodology used to produce the full provisional version of the DPCFam web app. Once on the global landing page, a user can choose DPCFam, PFam or UniproKB as the database, enter a valid ID, and then, if everything goes well :

- For DPCFam, they can explore Metacluster properties, download some files (Seeds, HMM model, MSA), and visit representatives(if any) in the AlphaFold database.
- For PFam, they can see a list of all matching DPCfam metaclusters and a box displaying the average overlap, and if desired, inspect the entered PFam ID in InterPro/Pfam database.
- For UniprotKB(proteins available in Uniref50), they can see a list of matching families in both DPCFam and Pfam. A visual graphic illustrating their overlaps(if any) across the full protein sequence length is displayed. If desired, they can inspect the full protein in UniProKB database.

All three databases sources are cross-linked in our project. A local PostgreSQL database is used for development purposes; relevant aspects will be discussed during our meeting.

# 2 Development

I explored the remaining DPCFam file : Uniref50\_annotated.xml.tar.gz, which is UniRef50 v.2017\_07 database annotated with Pfam families and DPCfam metaclusters. However, after unzipping it, I couldn't find the parser as started on Zenodo. What about a visual inspection of it (Figure 1)? :

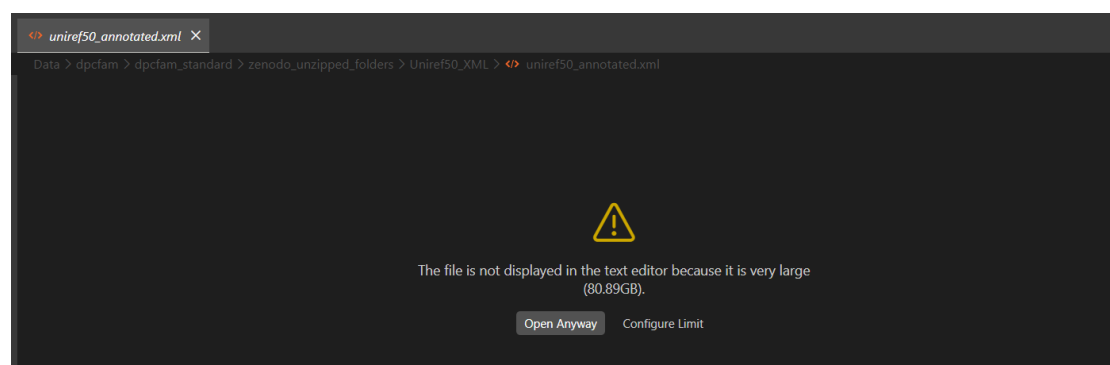


Figure 1: It is an 81 GB file.

Perhaps grepping a known protein ID could provide some insights(Figure 2):

Figure 2: Fortunately, the file contained what I wanted to see : Protein information.

[illegible]

We developed a notebook (available [here](#)) to preprocess the data from that `.csv` in a meaningful way, producing some dataframes(Figure 4):

|   | uniref50_id | uniprotkb_id | uniprotkb_accession | length |
|---|-------------|--------------|---------------------|--------|
| 0 | Q8WZ42-5    | Q8WZ42-5     | Q8WZ42-5            | 32900  |
| 1 | Q3ASY8      | Q3ASY8_CHLCH | Q3ASY8              | 36805  |
| 2 | G5B0U1      | G5B0U1_HETGA | G5B0U1              | 36507  |
| 3 | Q8WZ42      | TITIN_HUMAN  | Q9Y6L9              | 34350  |
| 4 | K7EE71      | K7EE71_ORNAN | K7EE71              | 7472   |

(a) Uniref50 - UniprotKB proteins.

(b) Uniref50 - Pfam

Figure 4: Uniref50 - Pfam information.

Then, we updated our database in PostgreSQL. Check out the current [README.md](#) for creating, optimising and populating the tables with data from `.csv` files.

We updated the project to include all tables. After connecting the database to Django, making migrations and running the server, choosing, for instance, DPCFam, MC1 on the landing page, the user can inspect an optimised version of MC1 tables(Figure 5):

| DPCFam & DPCStruct Unified Database                                                                                    |         |        |                                                                                        |
|------------------------------------------------------------------------------------------------------------------------|---------|--------|----------------------------------------------------------------------------------------|
| <a href="#">Home</a> <a href="#">About</a> <a href="#">Metaclusters</a> <a href="#">Downloads</a> <a href="#">FAQs</a> |         |        |                                                                                        |
| DPCFam   MC1                                                                                                           |         |        |                                                                                        |
| <a href="#">Summary</a> <a href="#">Sequences</a> <a href="#">3D Viewer</a> <a href="#">Downloads</a>                  |         |        |                                                                                        |
| Sequences                                                                                                              |         |        |                                                                                        |
| Protein ID                                                                                                             | Range   | Length | Amino Acids                                                                            |
| A0A0MOCOW3                                                                                                             | 3-192   | 190    | SITAVHSYKGGTGKTSLSVILAASLAKRGKVAZLDLDFRAPSLSFLKMHDSHLWINDYLGASTIHKILQETNGKGLNKGKLF...  |
| A0A1P8WPI3                                                                                                             | 533-701 | 169    | VRTAVCFNALGNAHRVQVTSFPAAGDGKSLAMRFVSLAQSGKCTLINESDFRRPKVHKLTVGVNKKGVVDVLRGKAEISDAIK... |
| A0A0P7YPQ9                                                                                                             | 114-308 | 195    | PGVRTTLAVSSGGGKGVKSTTAVNLAALQTGAKVGVLDADYGPVHPQHLGLGHADVEIVKTAQGRFKPLEAHGIQMSVGL...    |
| A0A1S9UN50                                                                                                             | 24-209  | 186    | RSKISEQYRSIRTNLQLSASTHNRTIVTSPRYGEGKSTTVNLAAASHAQKGEKVLVDANLRTPITIEHFQVNTTIGLTDILN...  |
| A0A094QKT1                                                                                                             | 2-116   | 115    | KTIAIASPAGVGKTTLAHAISVAAEFGKTLTLDLPAGSLTFRLGFENPRLTITDFLTGTSLGDENLNLTAEFAFIPADSR...    |
| C1F829                                                                                                                 | 251-474 | 224    | VPVLATFSVAGVGKTSLTANLGRALSSYGCPLLAETSTHGVLPFYFGARTPRPGVLRTFASHSAQEPVQHLSDGRHIAEG...    |
| A0A1Q5ZTP1                                                                                                             | 554-729 | 176    | EQIRSLRADLDFIPESKHNLIFTSISGEGKSFISLNLGASLASTGKKVIVLELDLRKPRFYITLGDHSVGLSSYLIGKQDY...   |
| W7CM65                                                                                                                 | 19-213  | 195    | VSEQIKAVRTSLNFATIDQDAQVILTSPTPSSGKSVSSNLAVSYAQDQKVLIIDGLRKATQHYQFKADPYRGLSSLIGKL...    |
| A0A0Q6JRN8                                                                                                             | 493-683 | 191    | ALRSALQLSTGAPKSLIFTSSGPAEGKSTSAIAIARFAQTIGHKVVLIADHRRPSQHRFEVONQKGLSNVLASQASLEOV...    |

Figure 5: DPCFam|MC1, Table2 : Sequences.

Choosing Pfam, plus a valid ID, results in something like Figure 6:

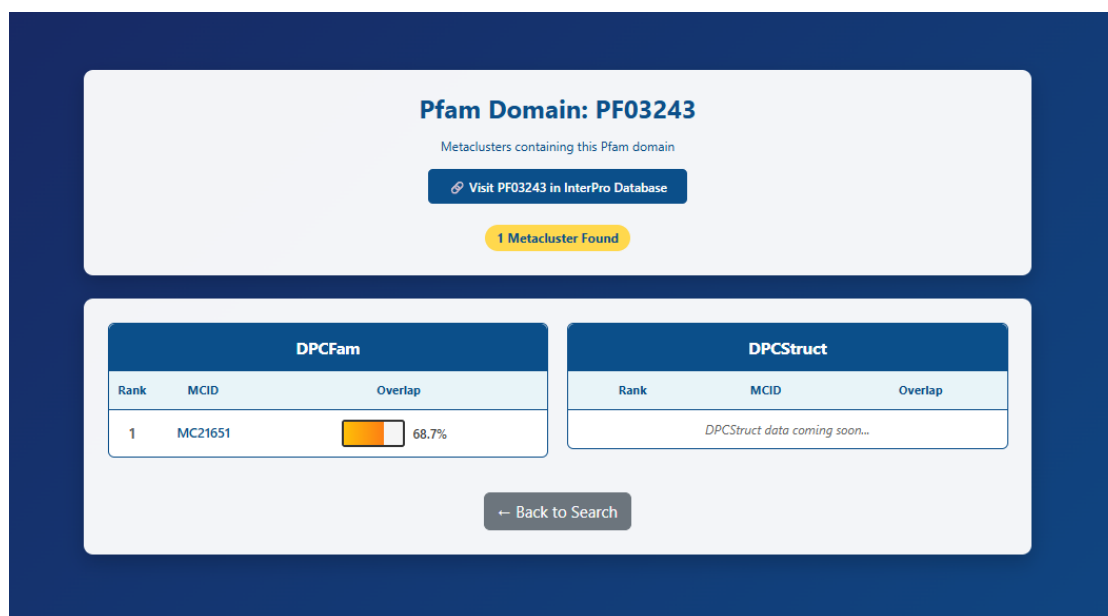


Figure 6: DPCFam MCs matching PF03243.

Choosing UniprotKB (Uniref50 protein IDs), with a valid protein ID results in a page like Figure 7:

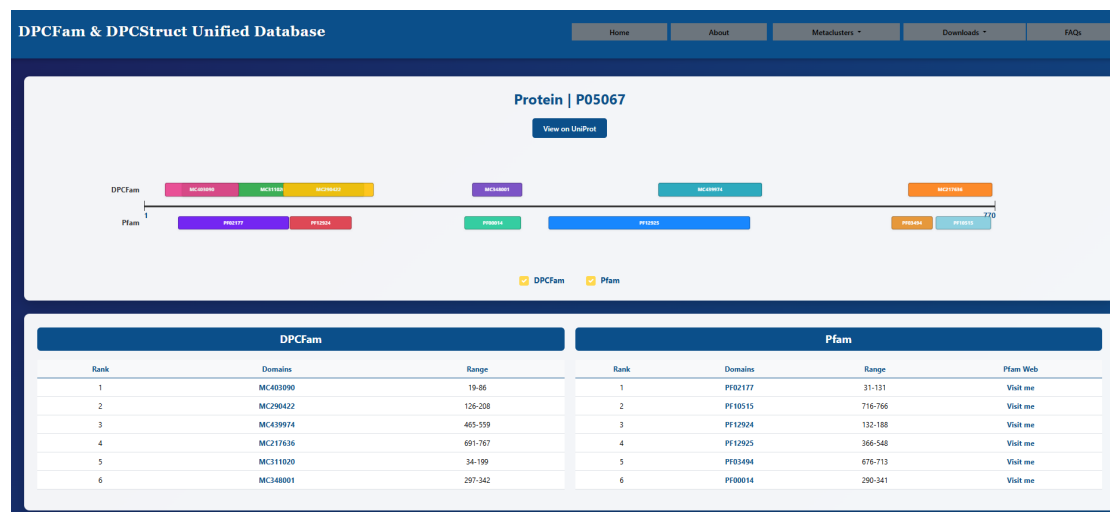


Figure 7: DPCFam MCs and PFam protein families matching protein P05067.

### 3 Conclusion

To display the current version of the project, a local PostgreSQL database configured as described in the README file on GitHub is required. This is a comprehensive version of the DPCFam app to be discussed during our upcoming meeting.

## References

- Russo, E. T., Barone, F., Bateman, A., Cozzini, S., Punta, M., and Laio, A. (2022). Dpccfam: unsupervised protein family classification by density peak clustering of large sequence datasets. *PLOS Computational Biology*, 18(10):e1010610.